III. CHI SQUARE TEST

The Chi square test for Goodness of Fit test claims about populations proportions.

It is non parametric test that is performed on categorical [ordinal and nominal] data.

- There is a population of male who like different color bikes.

	Theory	Sample
Yellow Bike	1/3	22
Red Bike	1/3	17
Orange Bike	1/3	59

* Goodness of fit test

Theory of categorical distribution

observed categorical distribution

Goodness of Fit test :

In a science class of 75 students, 11 are left handed. Does this class fit the theory that 12 % of people are left handed.

Ans.

	O	E
Left handed	11	9
Right handed	64	66
	75	75

$\frac{\cancel{12}^{3}}{\cancel{100}_{4}} \times \cancel{75}^{3} = 9$

O → observed

E → Expected,